Load testing – CargoSync

Products:

For testing product endpoint, we have focused on the most critical endpoints. These include:

- GET /api/categories
- GET /api/products
- GET /api/products?name=Cola&minPrice=50
- POST /api/products
- PUT /api/products/{id}

Category GET request is especially interesting since it has Redis cache. Product categories are very rarely changed, and are therefore and excellent candidate for caching.

Factories:

For factories, we chose these endpoints:

- GET /api/factories?cityCode=BEG
- GET /api/factories/{id}
- GET /api/factories/{id}/products/{id}/production

Search is not being cached, and therefore offers a constant challenge to the backend. Meanwhile the ID get is, and let us see the optimization achieved from the caching. Production endpoint is great since it needs to generate influx queries and is therefore heavy for the influx database and the communication between it and the backend.

Users:

Lastly for users the tested endpoints are:

- POST /api/auth/login
- GET /api/managers?query=manager555

Login let us see how difficult it is for the server to handle the constant password hashing.

Results:

| Scenario | Status | Requests | Failures | Fail % | RPS | P95 (ms) | Avg (ms) |

|---|---|---:|---:|---:|---:|---:|---:|

| products | PASS | 43 | 0 | 0 | 6.11 | 420 | 128.71 |

| factories | PASS | 56 | 0 | 0 | 8.07 | 78 | 26.62 |

| factory_analytics | PASS | 60 | 0 | 0 | 8.57 | 76 | 37.27 |

| login | PASS | 46 | 0 | 0 | 6.73 | 100 | 21.5 |